

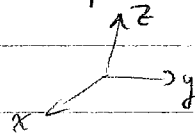
Heat Flux

$$q_x = -k \frac{\partial T}{\partial x}$$

$$q_y = -k \frac{\partial T}{\partial y}$$

$$q_z = -k \frac{\partial T}{\partial z}$$

In cartesian (rectangular) coordinate system:



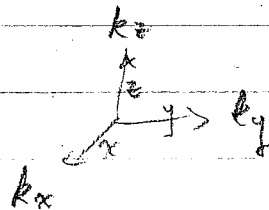
These expressions assume that k is independent of direction, i.e. material is isotropic.

In Orthotropic materials, k_x , k_y , & k_z are different.

$$q_x = -k_x \frac{\partial T}{\partial x}$$

$$q_y = -k_y \frac{\partial T}{\partial y}$$

$$q_z = -k_z \frac{\partial T}{\partial z}$$



ex wood, crystals, composites

crystal cut along natural grain (principal axis)

In Anisotropic materials, q depends on all three gradients

$$q_x = - \left[k_{11} \frac{\partial T}{\partial x} + k_{12} \frac{\partial T}{\partial y} + k_{13} \frac{\partial T}{\partial z} \right]$$

$$q_y = \dots$$

$$q_z = \dots$$

$$\underline{k} = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}$$

ex Crystal cut along arbitrary line

Some Amorphous materials.

or

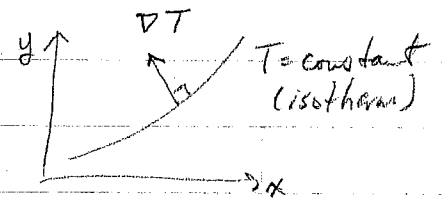
$$q_i = - \sum_j k_{ij} \frac{\partial T}{\partial x_j}$$

Heat Flux in vector form, the Fourier's Law

$$\vec{q} = -k \vec{\nabla} T$$

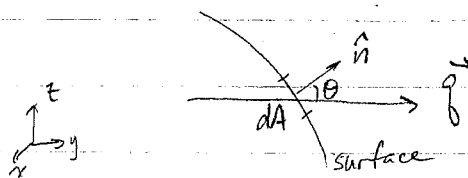
where $\vec{\nabla} T = \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right)$

$$= \hat{i} \frac{\partial T}{\partial x} + \hat{j} \frac{\partial T}{\partial y} + \hat{k} \frac{\partial T}{\partial z}$$



Heat flux is normal to isotherm.

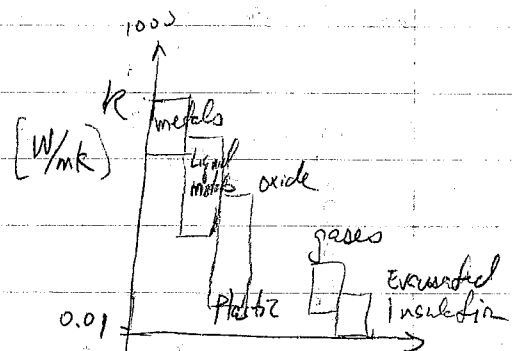
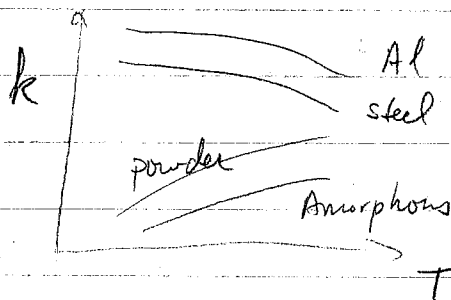
Material is isotropic $\Rightarrow k$ is independent of direction.



$$q_n = \vec{q} \cdot \hat{n} dA = |\vec{q}| |\hat{n}| dA \cos \theta$$

$$q_n = -k \frac{\partial T}{\partial n}$$

Thermal Conductivity (k)



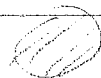
Best insulation for conduction HT is vacuum.

Diamond has $k \approx 700 \text{ W/mK}$, but has low electrical conduction.

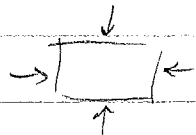
Formulation

Differential, Integral, or Lumped Formulation.

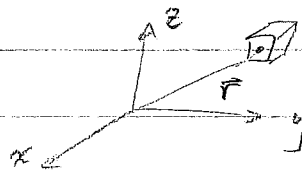
In Lumped formulation, material is independent of spatial variables.

 $T(t)$

Integral equations are expressed for a large finite region

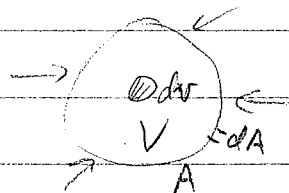


Differential Formulation applies equations on an infinitesimal volume



$T(\vec{r}, t)$

Use isotropic, homogeneous, stationary material.

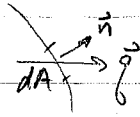


$$\left[\text{Heat coming in at the boundaries} \right] + \left[\text{Heat generated in the volume} \right] = \left[\begin{array}{l} \text{(or heat)} \\ \text{Energy stored} \\ \text{in the volume} \end{array} \right]$$

$\uparrow$
 $(q_{in} - q_{out})$

First term:

$$-\int_A \vec{q} \cdot \vec{n} dA$$



Negative sign appears because $\vec{n}$ is directed outward.

Using Divergence Theorem, $\int_A \vec{q} \cdot \vec{n} dA = \int_V \nabla \cdot \vec{q} dv$

$$-\int_A \vec{q} \cdot \vec{n} dA = -\int_V \nabla \cdot \vec{q} dv$$

Second term: $\int_V g dv$ where g has units $[W/m^3]$

$$\text{Third term: } \int_V \rho c_p \frac{\partial T}{\partial t} dv$$

For solids & liquids, $c_p \approx c_v \approx c$

For gases, use c_p OR c_v .

Combining terms.

$$\int_V [-\nabla \cdot \vec{q} + g - \rho c_p \frac{\partial T}{\partial t}] dv = 0$$

$$\int_V [\nabla \cdot (k \nabla T) + g - \rho c_p \frac{\partial T}{\partial t}] dv = 0$$

an integral equation. This equation allows us to get into finite volume or finite element methods.

Applying to small volume, dV becomes constant and can be factored out,

$$\boxed{\nabla \cdot (k \nabla T) + g = \rho c_p \frac{\partial T}{\partial t}}$$

We have a conduction equation for isotropic, stationary, homogeneous region with transience (Time dependence). $k, g, \rho c_p$ can depend on temperature or location.

for Steady State conditions, $\frac{\partial T}{\partial t} = 0$.

$$\nabla \cdot (k \nabla T) + g = 0$$

$g, \rho c_p$ drop out. Let $k = \text{constant}$.

$$\nabla^2 T + \frac{g}{k} = 0$$

a Poisson equation. If $g = 0$,

$$\nabla^2 T = 0$$

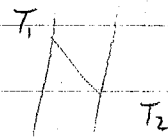
we have a Laplace equation.

In cartesian coordinates in 3-D,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$$

In 1-D, we have an ODE:

$$\frac{\partial^2 T}{\partial x^2} = 0$$



In transient with constant k ,

$$\nabla^2 T + \frac{q}{k} = \frac{\rho c_p}{k} \frac{\partial T}{\partial t}$$

Define thermal diffusivity $\alpha = \frac{k}{\rho c_p} \left[\frac{\text{m}^2}{\text{s}} \right]$ $\frac{[\frac{\text{W}}{\text{mK}}]}{[\frac{\text{kg}}{\text{m}^3}][\frac{\text{J}}{\text{kgK}}]}$
 α arises in transient problems.

It describes speed of propagation of thermal effects.

In cylindrical coordinates,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + q = \rho c_p \frac{\partial T}{\partial t}$$

Opening brackets
 & cancel out r may
 result in erroneous
 solutions!

In spherical coordinates,

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(k r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(k \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + q = \rho c_p \frac{\partial T}{\partial t}$$

General 2-D PDE

$$A \frac{\partial^2 \psi}{\partial x^2} + B \frac{\partial^2 \psi}{\partial x \partial y} + C \frac{\partial^2 \psi}{\partial y^2} + D(\psi, x, y, \frac{\partial \psi}{\partial x}, \frac{\partial \psi}{\partial y}) = 0.$$

Assume A, B, C are functions of x & y only (not ψ)

D is a linear function of ψ or no ψ dependence

$\Rightarrow$ We have a linear problem

$$\begin{aligned} \text{If } B^2 - 4AC < 0 & \text{ Elliptic} \\ & = 0 \text{ Parabolic} \\ & > 0 \text{ Hyperbolic} \end{aligned}$$

ex

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

$$0 - 4(1)(1) = -4 \Rightarrow \text{Elliptic}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$0 - 4(1)(0) = 0 \Rightarrow \text{Parabolic}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{\partial^2 T}{\partial t^2}$$

$$0 - 4(1)(-1) = 4 \Rightarrow \text{Hyperbolic}$$

In elliptic problems, boundary conditions effect interior everywhere. Sol'n depends on boundary everywhere!

Effect of parabolic problems is in one direction only. BC in future not needed.

Initial Condition

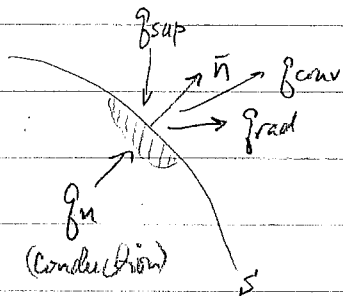
$$\text{At } t=0, T(\vec{r}, 0) = ?$$

Boundary Conditions

$$\text{for } t > 0, T(\vec{r}, t) = ?$$

$$x=0, T = \dots$$

$$x=1, T = \dots$$



$$q_{\text{conv}} = h(T - T_{\infty})$$

$$q_{\text{rad}} = \epsilon \sigma (T^4 - T_r^4)$$

$$q_n = -k \frac{\partial T}{\partial n} \quad \uparrow \text{effective ambient temperature}$$

For NC in air, $h = 5-10 \text{ W/m}^2\text{K}$

water, $h = 100-1000 \text{ W/m}^2\text{K}$

For Forced Conv. in air, $h = 40-100 \text{ W/m}^2\text{K}$

water, $h = 10^3-10^4 \text{ W/m}^2\text{K}$

In Boiling/Condensation, $h = 10^4-10^6 \text{ W/m}^2\text{K}$

Energy balance at the surface,

$$q_n + q_{\text{sup}} = q_{\text{conv}} + q_{\text{rad}}$$

$$-k \frac{\partial T}{\partial n} + q_{\text{sup}} = h(T - T_{\infty}) + \epsilon \sigma (T^4 - T_r^4)$$

A nonlinear eqn. with T unknown.

The radiation term can be expressed as $h_r(T-T_r)$ where $h_r = 4\epsilon\sigma T_r^3$ only if $T \approx T_r$. Higher order is more difficult to solve and may have multiple solutions!

Boundary Conditions

1st Kind

$$T = f(\vec{r}, t) \text{ on } S$$

$$T = 0 \quad \text{Homogeneous BC}$$

$$T = f(\vec{r}, t) \text{ on } S$$

Known also as Dirichlet Conditions, constant temperature.

2nd Kind

$$\frac{\partial T}{\partial n} = f(\vec{r}, t) \text{ on } S$$

$$\frac{\partial T}{\partial n} = 0 \quad \text{Adiabatic Condition (no energy transfer)}$$

Boundaries are insulated or symmetric

Known also as constant heat flux, Neumann Condition.

3rd Kind

$$A \frac{\partial T}{\partial n} + B T = f(\vec{r}, t) \text{ on } S$$

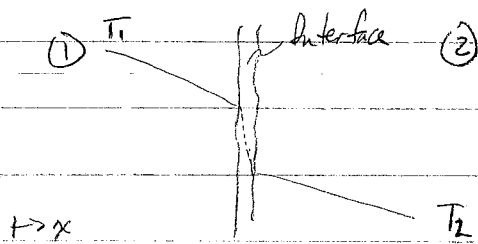
Mixed Conditions.

$$A \frac{\partial T}{\partial n} + B T = 0 \text{ is Homogeneous}$$

$$\rightarrow \text{for } -k \frac{\partial T}{\partial n} = h(T - T_\infty)$$

Known also as convective conditions.

Interface Condition



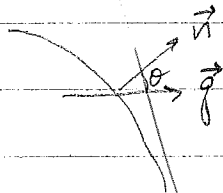
$$-k_1 \left. \frac{\partial T}{\partial x} \right|_1 = h_c (T_1 - T_2) = -k_2 \left. \frac{\partial T}{\partial x} \right|_2 \quad \text{for Steady State}$$

If contact is perfect, $T_1 = T_2$.

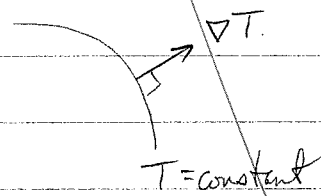
$$-k_1 \left. \frac{\partial T}{\partial x} \right|_1 = -k_2 \left. \frac{\partial T}{\partial x} \right|_2$$



$$\vec{q} = -k \nabla T$$



$$q_n = \vec{q} \cdot \vec{n} = -k \frac{\partial T}{\partial n}$$



S-S conduction : $\nabla^2 T = 0$

Transient variable property conduction : $\nabla \cdot (k \nabla T) + g = \rho c_p \frac{\partial T}{\partial \tau}$

Orthotropic :

$$\begin{cases} q_x = -k_x \frac{\partial T}{\partial x} \\ q_y = -k_y \frac{\partial T}{\partial y} \\ q_z = -k_z \frac{\partial T}{\partial z} \end{cases}$$

$$\frac{\partial}{\partial x} (k_x \frac{\partial T}{\partial x}) + \frac{\partial}{\partial y} (k_y \frac{\partial T}{\partial y}) + \frac{\partial}{\partial z} (k_z \frac{\partial T}{\partial z}) + g = \rho c_p \frac{\partial T}{\partial \tau}$$

Units

$$T = [K]$$

$$t = [s]$$

$$k = \left[\frac{W}{mK} \right]$$

$$g = [W/m^3]$$

$$\alpha = [m^2/s]$$

$$\rho = [kg/m^3]$$

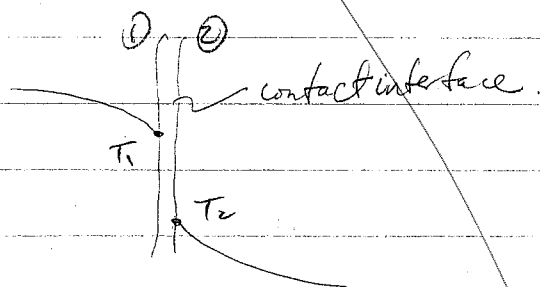
$$c_p = [J/kgK]$$

for k constant,

$$\nabla^2 T + \frac{q}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$$

where $\alpha = \frac{k}{\rho c_p}$ [m^2/s]

Contact Resistance



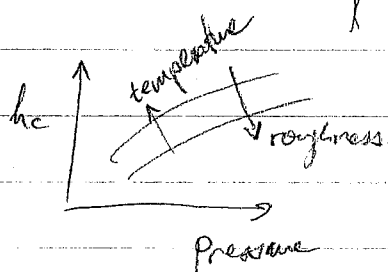
$-k_1 \frac{\partial T_1}{\partial x}|_i = -k_2 \frac{\partial T_2}{\partial x}$ at interface if contact is perfect $\Rightarrow T_2 = T_1$.

$-k_1 \frac{\partial T_1}{\partial x} = h_c(T_1 - T_2) = -k_2 \frac{\partial T_2}{\partial x}$ for imperfect contact $\Rightarrow T_2 \neq T_1$.

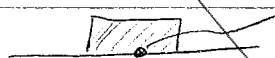
h_c due to roughness



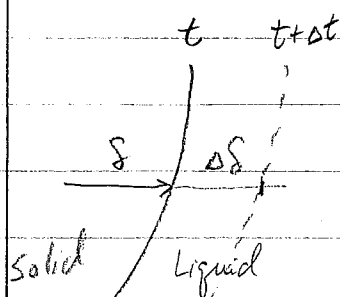
If roughness $\uparrow$, $h_c \downarrow$.
pressure $\uparrow$, $h_c \uparrow$.

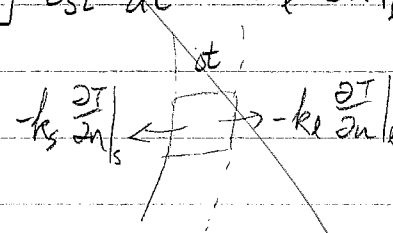


ex Chip bonding

 contact resistance governs cooling.

Solidification

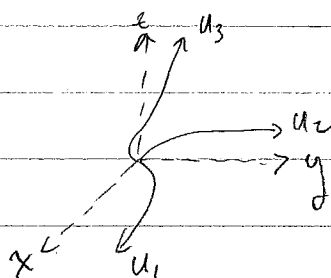


$$\rho L_{SL} \frac{ds}{dt} = -k_l \left. \frac{\partial T}{\partial n} \right|_{\text{liquid}}^{\text{out}} + k_s \left. \frac{\partial T}{\partial n} \right|_{\text{solid}}^{\text{in}}$$


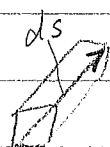
Latent heat $L_{SL} \left[\frac{J}{kg} \right]$

Coordinate Systems.

Curvilinear Orthogonal



$$\begin{aligned} x &= X(u_1, u_2, u_3) \\ y &= Y(u_1, u_2, u_3) \\ z &= Z(u_1, u_2, u_3) \end{aligned}$$



$$(ds)^2 = (dx)^2 + (dy)^2 + (dz)^2$$

or

$$dx = \sum_{i=1}^3 \frac{\partial x}{\partial u_i} du_i = \frac{\partial x}{\partial u_1} du_1 + \frac{\partial x}{\partial u_2} du_2 + \frac{\partial x}{\partial u_3} du_3$$

the chain rule for y & z ,

$$dy = \sum \frac{\partial y}{\partial u_i} du_i$$

$$dz = \sum \frac{\partial z}{\partial u_i} du_i$$

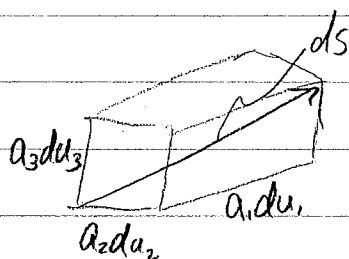
Insert dx, dy , & dz expressions in to $(ds)^2$ eqn,

$$(ds)^2 = a_1^2 (du_1)^2 + a_2^2 (du_2)^2 + a_3^2 (du_3)^2$$

where

$$a_i^2 = \left(\frac{\partial x}{\partial u_i}\right)^2 + \left(\frac{\partial y}{\partial u_i}\right)^2 + \left(\frac{\partial z}{\partial u_i}\right)^2 \text{ for } i=1,2,3$$

a_i are known as scaling factors



Since axes are normal, all cross derivatives will drop out.

The energy equation becomes

$$\nabla T = \hat{u}_1 \frac{1}{a_1} \frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{a_2} \frac{\partial T}{\partial u_2} + \hat{u}_3 \frac{1}{a_3} \frac{\partial T}{\partial u_3}$$

$$q_i = -k \frac{1}{a_i} \frac{\partial T}{\partial u_i}$$

Defining $a = a_1 a_2 a_3$, the equation becomes

$$\frac{1}{a} \left[\frac{\partial}{\partial u_1} \left(k \frac{a}{a_1^2} \frac{\partial T}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(k \frac{a}{a_2^2} \frac{\partial T}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left(k \frac{a}{a_3^2} \frac{\partial T}{\partial u_3} \right) \right] + \rho c_p \frac{\partial T}{\partial t}$$

$$dx \rightarrow a_1 du_1$$

$$dy \rightarrow a_2 du_2$$

$$dz \rightarrow a_3 du_3$$

for the case of cylindrical coordinates,

$$= x(r, \phi, z)$$

$$x = r \cos \phi, \quad y = r \sin \phi, \quad z = z$$

$$u_1 = r, \quad u_2 = \phi, \quad u_3 = z$$

Insert into expressions, $a_1 = 1$

$$a_2 = r$$

$$a_3 = 1$$

for spherical coordinates,

$$x = r \sin \theta \cos \phi \quad \text{with } r, \theta, \phi$$

$$y = r \sin \theta \sin \phi$$

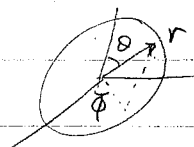
$$z = r \cos \theta$$

Calculating,

$$a_1 = 1$$

$$a_2 = r \sin \theta$$

$$a_3 = r$$



Coordinate Transforms can be used for complicated domains, i.e. ellipsoids, etc...

Orthogonal grids are preferred because axes are independent of one another, and are preferred over non orthogonal gridding.

Homogeneous

All terms go to zero.

ex

$$\nabla^2 T - \frac{1}{\alpha} \frac{\partial T}{\partial t} = 0$$

$$T=0 \text{ @ surface}$$

however,

$$T=T_s \text{ is not homogeneous. } \Rightarrow \theta = T - T_s$$

$\theta = 0 \text{ for } T = T_s$
transformed homogeneous.

A homogeneous linear eqn. multiplied by a constant is still the same answer.

$$-k \frac{\partial T}{\partial n} = h(T)$$

↓

$$k \frac{\partial T}{\partial n} + hT = 0 \Rightarrow \text{homogeneous.}$$

$$-k \frac{\partial T}{\partial n} = h(T - T_{\infty}) \quad \text{not homogeneous.}$$

transformed using $\theta = (T - T_{\infty})$.

$$-k \frac{\partial \theta}{\partial n} = h\theta \Rightarrow \text{homogeneous.}$$

$$\nabla^2 T + \frac{q}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{not homogeneous because of } \frac{q}{k} \text{ term}$$

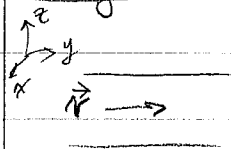
Split into

$$\nabla^2 T = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$\nabla^2 T + \frac{q}{k} = 0$$

} homogeneous.

Moving Solids



$$q_x = -k \frac{\partial T}{\partial x} + \rho c_p u_x T$$

The 1st term on RHS is conduction, the 2nd term on RHS is ~~convection~~ advection.
Similarly,

$$q_y = -k \frac{\partial T}{\partial y} + \rho c_p u_y T$$

$$q_z = -k \frac{\partial T}{\partial z} + \rho c_p u_z T$$

The energy equation becomes

$$\rho c_p \left[\frac{\partial T}{\partial t} + u_x \frac{\partial T}{\partial x} + u_y \frac{\partial T}{\partial y} + u_z \frac{\partial T}{\partial z} \right] = \nabla \cdot (k \nabla T) + \dot{q}$$

$$\rho c_p \left[\frac{\partial T}{\partial t} + \vec{v} \cdot \nabla T \right] = \nabla \cdot (k \nabla T) + \dot{q}$$

If $\vec{v} = 0$, we go back to the stationary solid.

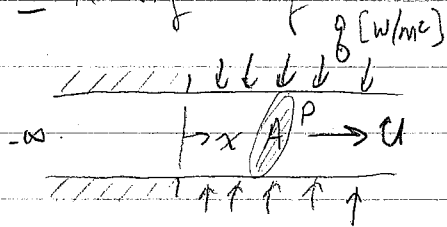
$\vec{v} \cdot \nabla T$: ~~convection~~ advection term

$\frac{\partial T}{\partial t}$: transient term

$\nabla \cdot (k \nabla T)$: diffusion term

$\dot{q}$: source term / generation term.

ex Heating moving rod.



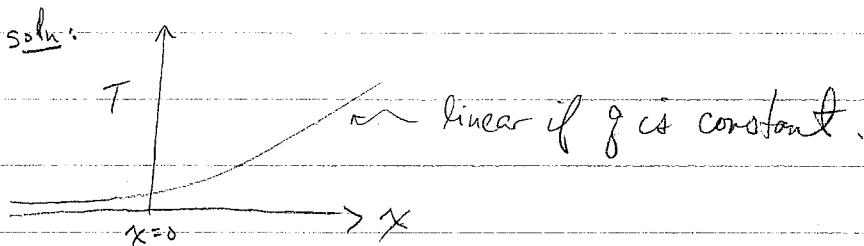
extrusion
hot rolling
drawing
pipe flows
permeate.

In heating section: $\frac{\rho C_p}{k} u \frac{dT}{dx} = \frac{d^2 T}{dx^2} + \frac{gP}{kA}$

In insulating section: $\frac{\rho C_p}{k} u \frac{dT}{dx} = k \frac{d^2 T}{dx^2}$

Both eqns are linear 2nd order ODE's.
2nd eqn is homogeneous.

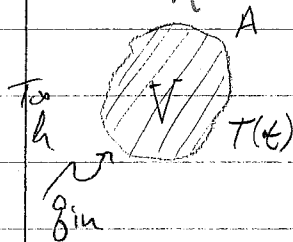
2nd order needs 2 BC's.



Graetz Problem: Uniform flow in a pipe.

Capacitance

Lumped Analysis (Method)



No spatial variation in temperature.

$$\dot{q}_{in} = h A (T_{\infty} - T)$$

$$\dot{q}_{stored} = \rho c_p V \frac{dT}{dt}$$

Energy balance:

$$h A (T_{\infty} - T) = \rho c_p V \frac{dT}{dt}$$

$$\rho c_p V \frac{dT}{dt} = -h A (T - T_{\infty}) \quad \text{losing energy}$$

IC: $t = 0, T = T_0$

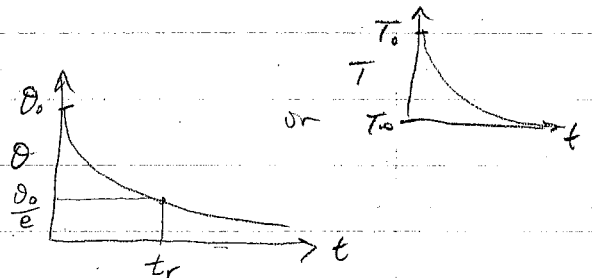
use $\theta = T - T_{\infty}$, $t = 0, \theta = T_0 - T_{\infty} = \theta_0$

$$\frac{d\theta}{dt} = -\frac{hA}{\rho c_p V} \theta$$

$$\theta = \theta_0 e^{\left[-\frac{hA}{\rho c_p V} t\right]}$$

At $t = 0, \theta = \theta_0$

As $t \rightarrow \infty, \theta \rightarrow 0$



Define the response time as

$$t_r = \frac{\rho c_p V}{hA}$$

with units [s]. When $t = t_r$, $\theta = \frac{\theta_0}{e}$.
Also known as e-folding time.

ex 1.5 in Özişik. Thermocouple in Flow.

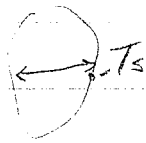
$$\frac{\theta}{\theta_0} = \frac{T - T_\infty}{T_0 - T_\infty} = \frac{1}{100} \text{ at } t = ?$$

$$\theta = \theta_0 e^{-\frac{hA}{\rho c_p V} t}$$

Inserting values, $t = 3.22$ seconds.

↓ radius results in ↓ t . (see t_r eqn.).

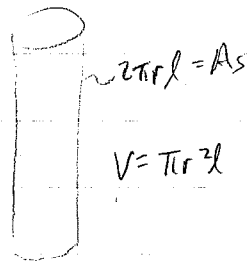
Validity of Lumped Assumption



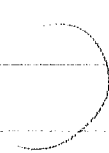
How is T inside material different from T_s .

Use Biot Number.

$$Bi = \frac{hL}{k_s} \quad \text{where } L = \frac{V}{A_s} \begin{matrix} \text{volume} \\ \text{surface area} \end{matrix}$$



$$L_c = \frac{V}{A_s} = \frac{r}{2}$$



$$V = \frac{4}{3} \pi r^3$$

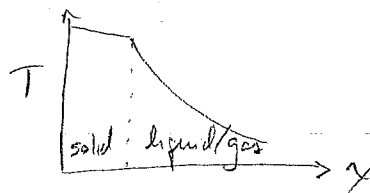
$$A_s = 4 \pi r^2$$

$$L_c = \frac{V}{A_s} = \frac{r}{3}$$

$$Bi = \frac{h}{k_s/L_c} = \frac{\text{conductive resistance}}{\text{convective resistance}} = \frac{\text{internal resistance}}{\text{external resistance}}$$

If Bi is small, most resistance is offered externally.

If $Bi < 0.1$, Lumped analysis is OK.



For $Bi < 0.1$, analysis is $< 5\%$.

Even for $Bi = 1$ or larger, lumped analysis can provide an estimated "ballpark" figure.
 ⇒ Very valuable assumption.

Heat Flux + Convective BC

$\delta \rightarrow T(t)$
 $A \quad V$

$$\rho C_p V \frac{\partial T}{\partial t} = \dot{q} A - hA (T - T_{\infty})$$

IC: $T = T_{\infty}$ at $t = 0$

Define $\theta = T - T_{\infty}$,

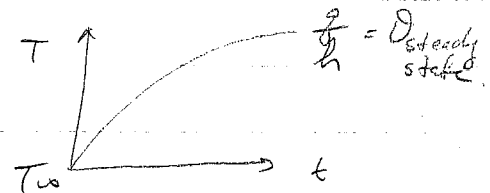
$$\frac{d\theta}{dt} = \frac{\dot{q} A}{\rho C_p V} - \frac{hA}{\rho C_p V} \theta$$

Solving,

$$\theta = \frac{\dot{q}}{h} \left[1 - e^{-\frac{hA}{\rho C_p V} t} \right]$$

At $t = 0$, $\theta = 0 \Rightarrow T = T_{\infty}$

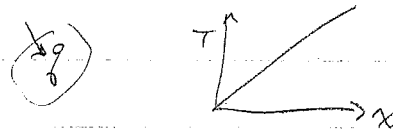
As $t \rightarrow \infty$, $\theta \rightarrow \frac{\dot{q}}{h}$



Heat Flux only

$$\rho C_p V \frac{\partial T}{\partial t} = \dot{q} A$$

$$T = \frac{\dot{q} A}{\rho C_p V} t$$



Not realistic

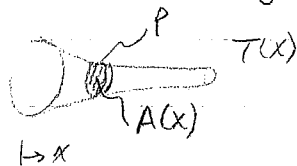
Radiation + Heat Flux

$$\rho C_p V \frac{dT}{dt} = qA - \epsilon \sigma (T^4 - T_{\infty}^4)$$

Nonlinear due to radiation term

Valid for $Bi < 0.1$

Partial Lumping - Extended Surface (Fin)



Look at small slice:

$$qA \xrightarrow[\text{+ } \Delta x]{P \Delta x h (T - T_{\infty})} qA + \frac{d(qA)}{dx} \Delta x$$

$$\underbrace{\frac{1}{A(x)} \frac{\partial}{\partial x} \left[A(x) \frac{\partial \theta}{\partial x} \right]}_{\text{in}} - \underbrace{\frac{hP(x)}{kA(x)} \theta(x,t)}_{\text{out}} = \underbrace{\frac{1}{\alpha} \frac{\partial \theta}{\partial t}}_{\text{storage}}$$

For $A_0 = \text{constant}$ ~~$A(x)$~~ ,

$$\frac{\partial^2 \theta}{\partial x^2} - \frac{hP}{kA_0} \theta(x,t) = \frac{1}{\alpha} \frac{\partial \theta}{\partial t}$$

for S.S. conditions,

$$\frac{d^2 \theta}{dx^2} = \frac{hP}{kA_0} \theta, \quad m^2 = \frac{hP}{kA_0}$$

Solution is

$$\theta = C_1 e^{mx} + C_2 e^{-mx}$$

$$a \theta = B_1 \cosh(mx) + B_2 \sinh(mx)$$

recall:

$$(D^2 - m^2)\theta = 0$$

$$D = \pm m$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

BC's



$\theta = \theta_L$ isothermal

$\frac{\partial \theta}{\partial x}|_L = 0$ adiabatic

$L \rightarrow \infty$ infinite

for an infinite fin,

$$\theta = \theta_0 e^{-mx}$$

energy in (Q) $\propto \frac{L}{T}$ is

$$Q = \sqrt{h P k A_0} \theta_0 = -k \frac{dT}{dx}|_{x=0}$$

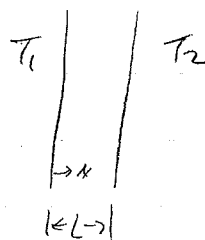
For an adiabatic end,

$$\theta = \theta_0 \frac{\cosh[m(L-x)]}{\cosh(mL)}$$

$$Q = \sqrt{hPkA_0} \theta_0 \tanh(mL)$$

As $L \rightarrow \infty$, $\tanh(\cdot) \rightarrow 1$

One-Dimensional S.S. Heat Conduction



$$\nabla^2 T = 0$$

$$\frac{d^2 T}{dx^2} = 0 \text{ for 1-D.}$$

$$\text{Soln: } T = C_1 x + C_2$$

$$T_1 = C_2$$

$$T_2 = C_1 L + C_2$$

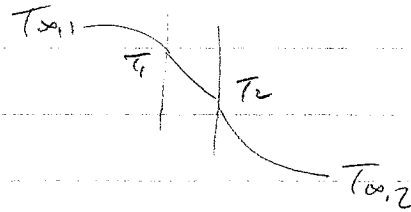
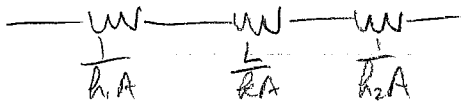
$$\Rightarrow T = T_1 + \frac{T_2 - T_1}{L} x$$

$$Q = \frac{T_1 - T_2}{\left(\frac{L}{kA}\right)}$$

$$Q = -kA \frac{\partial T}{\partial x} = -kA \frac{T_2 - T_1}{L}$$

$$Q = hA(T_2 - T_{\infty}) \left(\frac{L}{kA}\right) R_{\text{cond}}$$

$$= \frac{T_2 - T_{\infty}}{\left(\frac{1}{hA}\right)} R_{\text{conv.}}$$



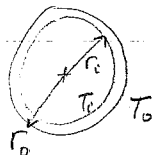
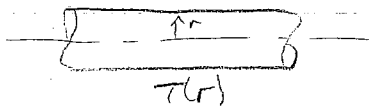
$$Q = \frac{T_{\infty,1} - T_{\infty,2}}{\frac{L}{k_1 A} + \frac{L}{k_2 A} + \frac{L}{k_3 A}}$$

ex



Parallel or series conduction

Cylinder



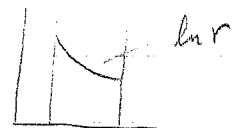
$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} = 0$$

Soln: $T = A \ln r + B$

Inserting BCs,

$$T = T_i - \frac{T_i - T_o}{\ln(r_o/r_i)} \ln(r/r_i)$$

where $R_{cond} = \frac{1}{2\pi k L} \ln(r_o/r_i)$



ex Pipe with insulation



Insulation thickness balances out with convective A

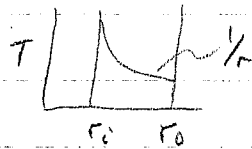
Spherical Case

$$\frac{d^2 T}{dr^2} + \frac{2}{r} \frac{dT}{dr} = 0$$

$$T = -\frac{C_1}{r} + C_2$$

$$R_{\text{cond}} = \frac{\frac{1}{r_i} - \frac{1}{r_o}}{4\pi k}$$

$$\theta = \frac{T_i - T_o}{R_{\text{cond}}}$$



Note: If we introduce transience in these eqns, we get pole's

Heat generation ($q \neq 0$).

T_0

$\left| \begin{array}{c} | \\ | \\ | - T_0 \\ | \\ k_s \\ | \\ L \\ | \\ \rightarrow x \end{array} \right.$

$\frac{d^2 r}{dx^2} + \frac{q}{k} = 0$

2nd order ODE.

dr

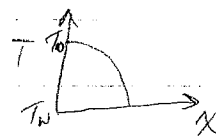
BC's: $x=0, \frac{dT}{dx}=0$ (symmetry)
 $x=L, T=T_w$

$$\frac{dz_T}{dx^2} = -\frac{g}{h}$$

$$\frac{dT}{dx} = -\frac{g}{h} x + C_1$$

$$T = -\frac{q}{k} \frac{x^2}{2} + C_1 x + C_2$$

$$T - T_w = \frac{q}{2h} (L^2 x^2)$$



Middle temperature T_0 is

$$T_0 - T_w = \frac{q}{2k} L^2$$

or

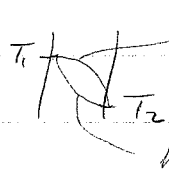
$$\frac{T - T_w}{T_0 - T_w} = 1 - \frac{x^2}{L^2}$$

Cylinder

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{q r}{k} = 0$$

Variable k $\rightarrow$ Results in nonlinearity.

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) = 0$$

 $k \uparrow$ with temperature.

$$k(T) \frac{dT}{dx} = C_1$$

$$k(T) dT = C_1 dx$$

Suppose $k(T) = BT$,

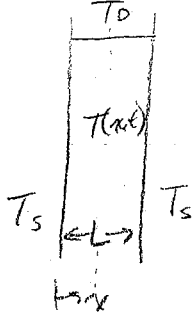
$$BT dT = C_1 dx$$

$$B \frac{T^2}{2} = C_1 x + C_2$$

Separation of Variables

class 1:

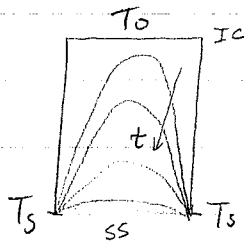
1-D transient



$$\text{IC } t=0: T=T_0 \text{ for } 0 \leq x \leq L$$

$$\text{BC } t>0: T=T_s \text{ @ } x=0, L \text{ or apply symmetry.}$$

We will get pde wrt x & t .



2-D transient: very tall

3-D transient: finite dimensions.

class 2: 2-D Steady State

$$T(x,y)$$

3-D Steady State

$$T(x,y,z)$$

Variables can be separated $T(x,t) = X(x)T(t)$

We won't know if separation of variables will work until we try it.

1-D transient

BC

$$\frac{\partial T}{\partial x} = 0 @ x=0$$

$\leftarrow L \rightarrow$

$\rightarrow x$

BC

$$-k \frac{\partial T}{\partial x} = hT @ x=L$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \text{ in region R}$$

$$T = F(x) @ t=0 \text{ for } 0 \leq x \leq L$$

Assume $T(x,t) = X(x)P(t)$

$$\underbrace{\frac{1}{X} \frac{d^2 X}{dx^2}}_{f(x) \text{ only}} = \underbrace{\frac{1}{\alpha} \frac{1}{P} \frac{dP}{dt}}_{f(t) \text{ only}} = -\beta^2$$

They can be equal only if they equal to a constant $(-\beta^2)$

$$\frac{dP}{dt} + \alpha \beta^2 P = 0$$

Soln: $P = e^{-\alpha \beta^2 t}$

We choose $-\beta^2$ as constant because we want time to decay.

$$\frac{d^2 X}{dx^2} + \beta^2 X = 0$$

BC's:

$$\frac{dX}{dx} = 0 @ x=0$$

$$k \frac{\partial X}{\partial x} + hX = 0$$

This is known as an eigenvalue problem. It can be satisfied by certain values of β .

If $\beta \neq \beta_m$, $X=0$ Trivial Solution, else we get $X(\beta_m, x)$ eigen functions.

We have a whole series of $\beta_m \neq X$.
This is known as linear superposition.
Solving, we get

$$T(x,t) = \sum_{m=1}^{\infty} C_m X(\beta_m, x) e^{-\alpha \beta_m^2 t}$$

a series solution. Once we obtain $T(x,t)$, we need to satisfy the IC

$$F(x) = \sum_{m=1}^{\infty} C_m X(\beta_m, x)$$

Orthogonality of $X(\beta_m, x)$ is used to evaluate $F(x)$.

$$\int_0^L X(\beta_m, x) X(\beta_n, x) dx = 0 \quad \text{if } m \neq n.$$

If $m=n$,

$$\int_0^L X(\beta_m, x) X(\beta_m, x) dx = N(\beta_m)$$

an normalization function.

$$\int_0^L [X(\beta_m, x)]^2 dx = N(\beta_m)$$

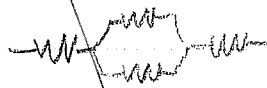
$$C_m = \frac{1}{N} \int_0^L X(\beta_m, x) F(x) dx.$$

We get a series solution.

Talked about 1-D, S.S. Elementary ODEs:

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

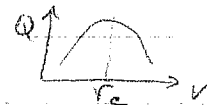
Looked at problems with & without q.
Resistances were defined as



Critical Radius

$\uparrow r$ results in $\downarrow$ convective resistance &
 $\uparrow$ conductive resistance.

There is an optimum r value:



Looked at 1-D transient problems. $T(x, t)$

Looked at lumped analysis. $T(t)$

1-D transient

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$\left. \frac{\partial T}{\partial x} = 0 \right|_{x=0, t>0} \quad \xrightarrow{L \rightarrow x} \quad \left. k \frac{\partial T}{\partial x} + hT = 0 \right|_{x=L, t>0}$$

Use $\theta = T - T_{\infty}$ to transform
BC's into homogeneous
problem

IC: $T = F(x)$ @ $t=0$ for $0 \leq x \leq L$.

Look at BC: (Mixed Condition or 3rd kind)

$$k \frac{\partial T}{\partial x} + hT = 0$$

If $h \rightarrow 0$, $\frac{\partial T}{\partial x} = 0$ adiabatic

$h \rightarrow \infty$, $T \rightarrow 0$

$$\begin{array}{c|c} \rightarrow & h(T - T_\infty) \\ -k \frac{\partial T}{\partial x} & \rightarrow \end{array}$$

$$-k \frac{\partial T}{\partial x} = h(T - T_\infty)$$

Defining $\theta = T - T_\infty$,

$$k \frac{\partial \theta}{\partial x} + h\theta = 0$$

Separation of Variables

Convert PDE $\rightarrow$ ODE

Assume $T(x, t) = X(x) \Gamma(t)$

This requires homogeneous linear BCs.

Insert into pde,

$$\frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{\Gamma} \frac{d\Gamma}{dt} = -\beta^2$$

Solving for Γ ,

$$\Gamma = e^{-\beta^2 t}$$

For X ,

$$\frac{d^2 X}{dx^2} + \beta^2 X = 0$$

with BCs

$$\frac{dX}{dx} = 0 \quad @ \quad x=0$$

$$k \frac{dX}{dx} + hX = 0 \quad @ \quad x=L$$

β can only pick up certain values, i.e. β_m eigenvalues.
Corresponding to eigenvalues, we also have eigenfunctions
 $X(\beta_m, x)$.

If plate is of finite thickness, we have discrete β_m ⁽¹⁾
If plate is infinite/semi-infinite, β becomes continuous.

Solution in general form is a series solution:

$$T(x,t) = \sum_{m=1}^{\infty} C_m X(\beta_m, x) e^{-\alpha \beta_m^2 t}$$

Applying IC,

$$F(x) = \sum_{m=1}^{\infty} C_m X(\beta_m, x)$$

Use orthogonality relation: $\int_0^L X(\beta_m, x) X(\beta_n, x) dx = \begin{cases} 0 & \text{if } m \neq n \\ N(\beta_m) & \text{if } m = n \end{cases}$

where

$$N(\beta_m) = \int_0^L X^2(\beta_m, x) dx$$

The constant is evaluated using

$$C_m = \frac{1}{N(\beta_m)} \int_0^L X(\beta_m, x) F(x) dx$$

In a series solution, most cases require only first couple of terms.

ex

$F(x)$

$$(-k_1 \frac{\partial T}{\partial x} + h_1 T = 0)$$

$$k_2 \frac{\partial T}{\partial x} + h_2 T = 0$$

material is long
heat convection on
both sides.

x

Assume $T(x,t) = X(x) \Gamma(t)$

Solving, $\Gamma(t) = e^{-\alpha \beta^2 t}$

$$\frac{d^2 X}{dx^2} + \beta^2 X = 0$$

$$(D^2 + \beta^2)X = 0$$

$$D = \pm i\beta$$

$$X = A_1 \cos \beta x + A_2 \sin \beta x$$

$$-k_1 \frac{\partial X}{\partial x} + h_1 X = 0 \quad @ \quad x=0$$

$$k_2 \frac{\partial X}{\partial x} + h_2 X = 0 \quad @ \quad x=L$$

Apply BC's,

$$-k_1 [-A_1 \beta \sin \beta x + A_2 \beta \cos \beta x]_{x=0} + h_1 (A_1 \cos \beta x + A_2 \sin \beta x)_{x=0} = 0$$

$$k_1 A_2 \beta = h_1 A_1$$

$$A_2 \beta = H_1 A_1 \quad \text{where } H_1 = \frac{h_1}{k_1}$$

Similarly,

$$H_2 = \frac{h_2}{k_2}$$

Applying BCs @ $x=L$, & combining signs,

$$\tan \beta L = \frac{(H_1 + H_2) \beta}{\beta^2 - H_1 H_2}$$

if $H_1 \neq 0$

Solving for β gives eigenvalues β_m , $m=1, 2, 3, \dots$

or

Making $h_1 \rightarrow 0$, we have adiabatic conditions @ $x=0$ ($H_1=0$)

Making $h_2 \rightarrow \infty$, we have isothermal conditions @ $x=L$.

$$\rightarrow \left. \frac{\partial T}{\partial x} = 0 \right| \left. T = 0 \right|$$

General solution form is

$$T(x, t) = \sum_{m=1}^{\infty} C_m X(\beta_m, x) e^{-\alpha \beta_m^2 t}$$

where

$$X = \beta_m \cos \beta_m x + H_1 \sin \beta_m x$$

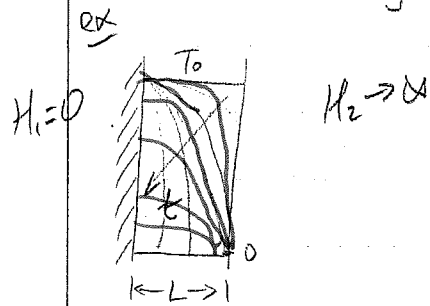
$$N(\beta_m) = \int_0^L X^2 dx$$

$$= \frac{1}{2} \left[(\beta_m^2 + H_1^2) \left(L + \frac{H_2}{\beta_m^2 + H_2} \right) + H_1 \right]$$

$$C_m = \frac{1}{N(\beta_m)} \int_0^L X(\beta_m, x) F(x') dx'$$

β depends on BC's.

$T(x,t)$ depends on time (t), space (x), material (α), and boundary conditions (β).



$$T = \underbrace{e^{-\alpha \beta^2 t}}_{P(t)} \underbrace{(A \cos \beta x + A_2 \sin \beta x)}_{X(\beta, x)}$$

BC's give $\cos \beta L = 0 \Rightarrow \beta_m L = (2m+1) \frac{\pi}{2}, m = 0, 1, 2, 3, \dots$
 $\text{or } (2m-1) \frac{\pi}{2}, m = 1, 2, 3, \dots$

IC's:

$$T_0 = \sum_{m=0}^{\infty} C_m \cos \beta_m x$$

Mult. by $\cos \beta_m x$ & integrate $0 \rightarrow L$,

$$C_m = (-1)^m \frac{2T_0}{\beta_m L}$$

We get

$$T(x,t) = 2T_0 \sum_{m=0}^{\infty} (-1)^m \frac{e^{-\alpha \beta_m^2 t}}{\beta_m L} \cos \beta_m x$$

ex

$$\frac{\partial T}{\partial x} = 0$$

$$k \frac{\partial T}{\partial x} + h_2 T = 0$$

$$H_1 = 0$$

$$H_2 = H$$

Solving for eigenvalues,

$$\tan \beta_m L = \frac{y}{\beta_m} = \frac{h}{k \beta_m} = \frac{hL}{k(\beta_m L)}$$

$$\tan \beta_m L = \frac{Bi}{\beta_m L}$$

where $Bi = \frac{hL}{k}$. Eigenvalues depend on Bi .

Solution is

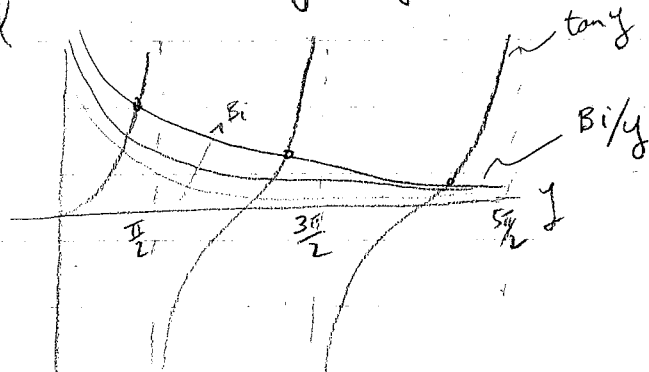
$$T(x,t) = \sum_{m=1}^{\infty} C_m e^{-\alpha \beta_m^2 t} \cos \beta_m x$$

$$C_m = 2T_0 \frac{1}{\cos \beta_m L} \frac{H}{L[\beta_m^2 + H^2] + H}$$

Look at form

$$\tan y = \frac{Bi}{y}$$

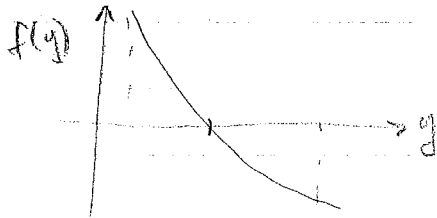
Graphical



Numerical Methods in Root Solving

$$\tan y = \frac{Bi}{y}$$

Write in form $f(y) = 0$

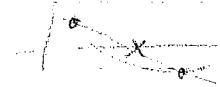


- Bisection - Break interval in half & isolate root location.
Slower.

- Newton Raphson - Use gradient to approximate root location.
Very fast.

- Secant - Similar to Newton Raphson.

Start with 2 points & find intersection



Final Solution

The answer is in the form,

$$T(x,t) = f(x, L, t, Bi, \alpha)$$

Nondim,

$$\frac{T}{T_0} = f_n\left(\frac{x}{L}, Bi, \frac{\alpha t}{L^2}\right)$$

Where $Fo = \frac{\alpha t}{L^2}$, Fourier number $\left(\frac{\text{rate of conduction in}}{\text{energy storage}} \right)$ representing nondim. time.

Nondim eqns

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

use $x^* = \frac{x}{L}$

$$Fo = \frac{\alpha L^2}{L^2}$$

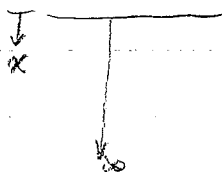
$$T^* = \frac{T - T_0}{T_0 - T_0}$$

$$Bi = \frac{hL}{k} \leftarrow \text{appears in BC's.}$$

$$\frac{\partial^2 T^*}{\partial x^{*2}} = \frac{1}{Fo} \frac{\partial T^*}{\partial t^*}$$

Semi-infinite Plates

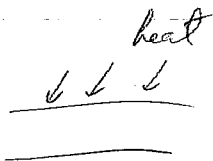
$x=0$



ex: Earth - No change @ $x > 10 \text{ m} \Rightarrow$ Shallow ponds
 $\alpha \approx 10^{-6} \text{ m}^2/\text{s}$

ex: Plastic

Low k .



Very low penetration. ($\sqrt{\alpha t}$ small)

$\Rightarrow$ bottom boundary can be removed & problem treated as nondim.

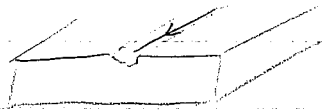
We have a "thick plate" or "short time" problem.
 Penetration depth δ is

$$\delta \propto \sqrt{\alpha t} \quad [\text{length}]$$

If $\frac{L}{\sqrt{\alpha t}} \gg 1 \Rightarrow \text{Semi-infinite}$

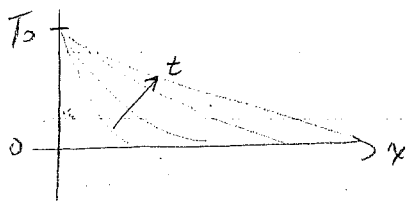
ex Laser heat

Laser exists for fractions ($\mu\text{s} \sim \text{ns}$) of seconds.

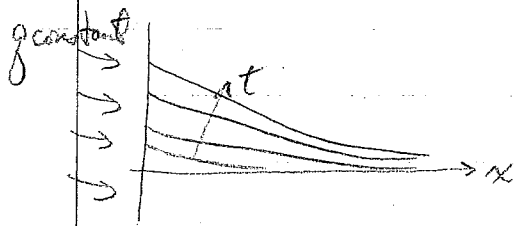


But time scale is so small that material can be treated as semi-infinite.

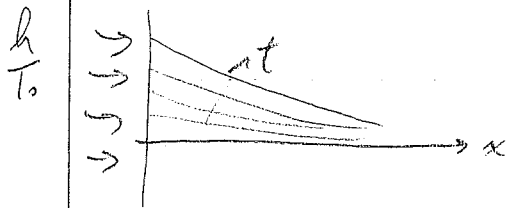
Boundaries



Temperature will not reach T_0 because material is semi-infinite.



Temperature can keep on rising ($\rightarrow \infty$).



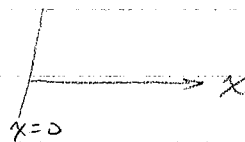
Temperature is limited by T_0 .

For a short time, everything behaves like an semi-infinite.

Egn

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

BC: $-k_1 \frac{\partial T}{\partial x} + h_1 T = 0 \quad @ \quad x=0$



IC: $T = F(x) \quad @ \quad t=0$

Assume $T(x,t) = X(x) \Gamma(t)$

Soln: $\Gamma(t) = e^{-\alpha \beta^2 t}$

$$\frac{d^2 X}{dx^2} + \beta^2 X = 0.$$

$$X(\beta, x) = \beta \cos \beta x + H_1 \sin \beta x, \quad H_1 = \frac{h_1}{\beta}$$

Notice that $0 \leq \beta \leq \infty$ is a continuous variation.

$$T(x,t) = \int_{\beta=0}^{\infty} C(\beta) e^{-\alpha \beta^2 t} (\beta \cos \beta x + H_1 \sin \beta x) d\beta$$

$$C(\beta) = \frac{1}{N(\beta)} \int_0^{\infty} X(\beta, x') F(x') dx'$$

$$\frac{1}{N(\beta)} = \frac{2}{\pi} \frac{1}{\beta^2 + H_1^2}$$

ex 23

If at $x=0$, $\frac{dX}{dx} = 0$, (adiabatic)

$$X = \cos \beta x$$

$$\frac{1}{N(\beta)} = \frac{2}{\pi}$$

If at $x=0$, $X=0$, (isothermal).

$$X = \sin \beta x$$

$$\frac{1}{N(\beta)} = \frac{2}{\pi}$$

And $F(x)$ is defined $\begin{array}{c} T(x) \\ \xrightarrow{\quad} x \end{array} \quad F(x) \quad \& \quad T(x=0, t) = 0$

We get

$$T(x, t) = \frac{2}{\pi} \int_0^{\infty} F(x') \int_{-\infty}^{\infty} e^{-\alpha \beta^2 t} \sin \beta x \sin \beta x' d\beta dx'$$

Recall:

$$\sin \beta x \sin \beta x' = \cos \beta (x-x') - \cos \beta (x+x')$$

From integral tables:

$$\int_0^{\infty} e^{-\alpha \beta^2 t} \cos \beta (x-x') d\beta = \sqrt{\frac{\pi}{4\alpha t}} \exp\left[-\frac{(x-x')^2}{4\alpha t}\right]$$

Use transformations to simplify integral.

Final Result

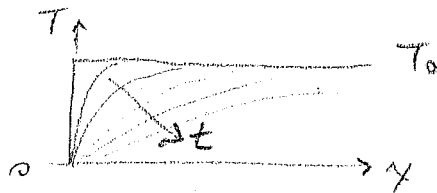
If $F(x) = T_0$,

$$\frac{T(x,t)}{T_0} = \frac{2}{\pi} \int_0^{\frac{x}{\sqrt{4\alpha t}}} e^{-\xi^2} d\xi$$

$$\boxed{\frac{T}{T_0} = \text{erf}(\eta)} \quad \text{where} \quad \eta = \frac{x}{\sqrt{4\alpha t}} = \frac{x}{2\sqrt{\alpha t}}$$

As $\eta = 0$, $\text{erf}(\eta) = 0$

As $\eta \rightarrow \infty$, $\text{erf}(\eta) \rightarrow 1$



$\eta = 2-3$ is enough to make $\text{erf}(\eta) \rightarrow 1$.

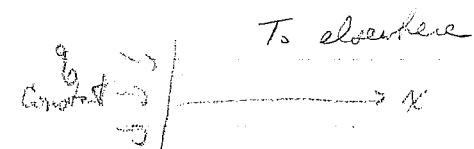
Flux Formulation

Insert $q = -k \frac{\partial T}{\partial x}$

into $\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

we get

$$\frac{\partial^2 q}{\partial x^2} = \frac{1}{\alpha} \frac{\partial q}{\partial t}$$

Define $q = f_0 = \text{constant @ } x=0$  $\xrightarrow{x}$

Use $Q = q - f_0$,

$$\frac{\partial^2 Q}{\partial x^2} = \frac{1}{\alpha} \frac{\partial Q}{\partial t}$$

BC: $Q = -f_0 @ x=0$.

Soln: $Q = -f_0 \operatorname{erf}\left(\frac{x}{\sqrt{4\alpha t}}\right)$

$$q = f_0 + Q = f_0 \left[1 - \operatorname{erf}\left(\frac{x}{\sqrt{4\alpha t}}\right) \right]$$

$$q = f_0 \operatorname{erfc}\left(\frac{x}{\sqrt{4\alpha t}}\right)$$

Temperature is found using $q = -k \frac{\partial T}{\partial x}$,

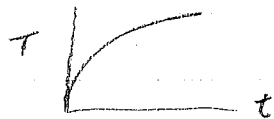
$$T(x,t) = \frac{f_0}{k} \int_0^x \operatorname{erfc}\left(\frac{x'}{\sqrt{4\alpha t}}\right) dx'$$

from appendix III,

$$T(x,t) = \frac{2f_0}{k} \left[\left(\frac{\alpha t}{\pi}\right)^{1/2} e^{-\frac{x^2}{4\alpha t}} - \frac{x}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{4\alpha t}}\right) \right]$$

At $x=0$,

$$T(0,t) = \frac{2f_0}{k} \left(\frac{\alpha t}{\pi}\right)^{1/2} \alpha t^{1/2}$$

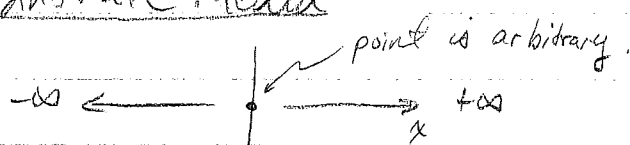


Temperature rises to $t^{1/2}$

This result is used to start constant q problems.

Notice that $\frac{\partial T}{\partial t}|_{t=0} = \infty$!! Codes will start with some simple analytical solution to avoid singularity.

Infinite Media



Glass heating with a laser beam:



$L \gg \sqrt{4\alpha t} \Rightarrow$ treat as infinite

Integrals are similar to sem infinite, by limits are $-\infty \rightarrow +\infty$.

eqn:
$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

No boundary conditions.

IC: $T = f(x) @ t = 0$

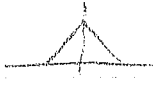
Soln:

$$T(x,t) = \frac{1}{\sqrt{4\pi\alpha t}} \int_{-\infty}^{\infty} F(x') \exp\left[-\frac{(x-x')^2}{4\alpha t}\right] dx'$$

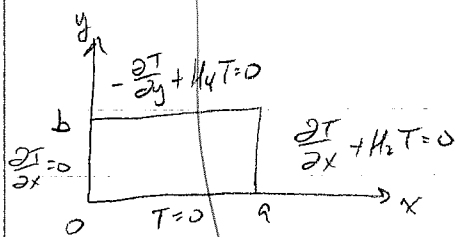
ex 2-5

$F(x) = \begin{cases} T_0 & \text{for } -L \leq x \leq L \\ 0 & \text{elsewhere} \end{cases}$

$$\begin{aligned} T(x,t) &= \frac{T_0}{\sqrt{4\pi\alpha t}} \int_{-L}^L \exp\left[-\frac{(x-x')^2}{4\alpha t}\right] dx' \\ &= \frac{1}{2} \left[\operatorname{erf} \frac{L+x}{\sqrt{4\alpha t}} + \operatorname{erf} \frac{L-x}{\sqrt{4\alpha t}} \right] \end{aligned}$$

$F(x)$ can also be , or many other cases.

Multidimensional Problems



$T = F(x, y)$ at $t=0$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

Assume $T(x, y, t) = X(x) Y(y) P(t)$

$$\underbrace{\frac{1}{X} \frac{d^2 X}{dx^2}}_{(1)} + \underbrace{\frac{1}{Y} \frac{d^2 Y}{dy^2}}_{(2)} = \frac{1}{\alpha} \frac{dP}{dt} = -A^2$$

Terms ① & ② are constants also, but they don't need to be the same constant.

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -\beta^2$$

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = -\gamma^2$$

$$A^2 = \beta^2 + \gamma^2$$

Soln: $T = e^{-\alpha(\beta^2 + \gamma^2)t}$

$$\frac{d^2 X}{dx^2} + \beta^2 X = 0$$

$$\frac{d^2 Y}{dy^2} + \gamma^2 Y = 0.$$

X & Y eqns will each give rise to a set of eigenvalues and eigen functions.

$$T(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} C_{mn} e^{-\alpha(\beta_m^2 + \gamma_n^2)t} X(\beta_m, x) Y(\gamma_n, y)$$

Solutions can be determined as long as BC's are homogeneous & $F(x, y)$ only.

$$C_{mn} = \frac{1}{N(\beta_m) N(\gamma_n)} \int_{x'=0}^a \int_{y'=0}^b X(\beta_m, x') Y(\gamma_n, y') F(x', y') dx' dy'$$

$$N(\beta_m) = \int_0^a X^2 dx$$

$$N(\gamma_n) = \int_0^b Y^2 dy$$

for this particular problem,

$$X = \cos(\beta_m x)$$

$$Y = \sin(\gamma_n y)$$

which satisfies 1-D BC's done before.

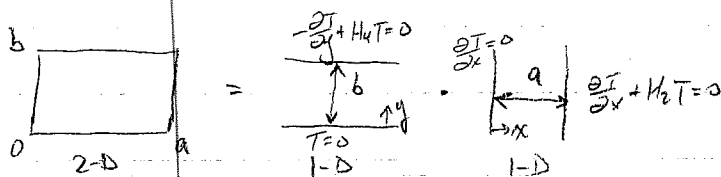
$$\beta_m \tan \beta_m a = H_2 \quad \text{If } H_2 \rightarrow \infty, \beta_m a \rightarrow \frac{n\pi}{2}$$

$$\gamma_n \tan \gamma_n b = -H_4$$

$$\frac{1}{N(\beta_m)} = 2 \frac{\beta_m^2 + H_2^2}{a(\beta_m^2 + H_2^2) + H_2}$$

$$\frac{1}{N(\gamma_n)} = 2 \frac{\gamma_n^2 + H_4^2}{b(\gamma_n^2 + H_4^2) + H_4}$$

Product Solutions



$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \frac{\partial^2 T}{\partial y^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

Say $T = T_1(x,t) \cdot T_2(y,t)$. Insert into 2-D pde.

$$T_2 \frac{\partial^2 T_1}{\partial x^2} + T_1 \frac{\partial^2 T_2}{\partial y^2} = \frac{1}{\alpha} \left[\frac{\partial T_1}{\partial t} T_2 + \frac{\partial T_2}{\partial t} T_1 \right]$$

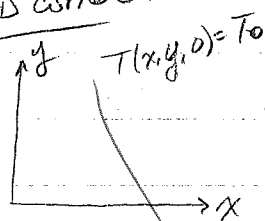
③ ④

③ & ④ satisfy both 1-D pde's simultaneously.

This argument also applies to BC's.

Product Solutions work if eqn. & BC's are homogeneous.

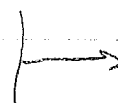
ex 2-D corner.



$$T(x, y, 0) = T_0$$

for short
time

a. semi-infinite

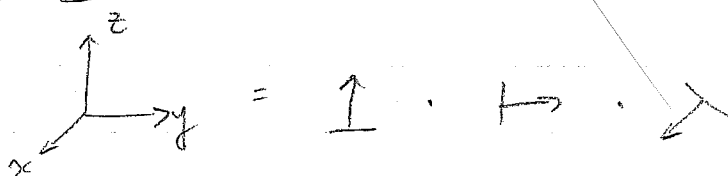


$$T(x, y, t) = T_0 \left(\operatorname{erf} \frac{x}{\sqrt{4\alpha t}} \right) \cdot \left(\operatorname{erf} \frac{y}{\sqrt{4\alpha t}} \right)$$

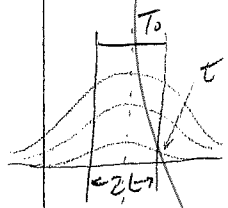
Initial conditions $F(x, y) = F_1(x) F_2(y)$ must be satisfied.

Suppose $T_0 = 1 \cdot T_0 \Rightarrow$ separated, OK.

ex 3-D corner



Charts - 1-D Transient



h, T_{∞} similar to



$$\begin{aligned} h, T_{\infty} \\ \theta = T - T_{\infty} \\ k \frac{\partial \theta}{\partial x} + h\theta = 0 \end{aligned}$$

$$\tan \beta_m L = \frac{\beta_m}{\beta_m L}$$

a transcendental eqn solved numerically.

Heisler Charts

$$\begin{aligned} Bi &= \frac{hL}{k} \\ Fo &= \frac{\alpha t}{L^2} \end{aligned}$$

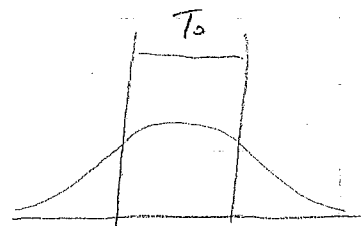
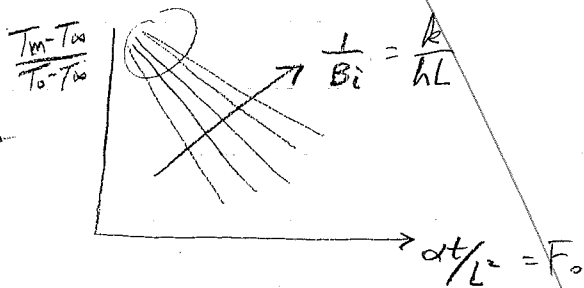
$$x^* = \frac{x}{L}$$

T_m = midplane temperature.

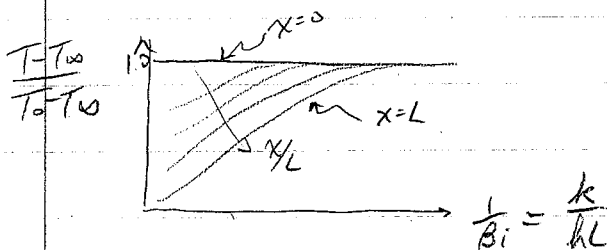
T_0 = initial temperature.

$$\frac{T - T_{\infty}}{T_0 - T_{\infty}} = f\left(\frac{x}{L}, Bi, Fo\right)$$

$$T_m \sim f(Fo, Bi)$$



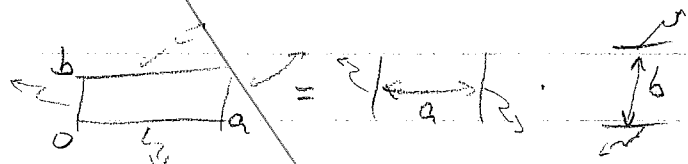
Heisler Charts give inaccurate solutions at $Fo \leq 1$ since it uses 1st term only.



$$\frac{T}{T_m} \sim f\left(Bi, \frac{x}{L}\right)$$

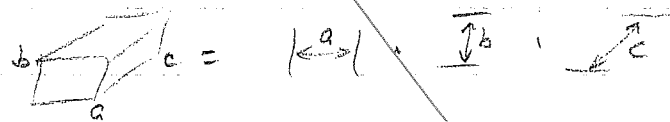
for small Bi , temperature differences are very small.
 $L \rightarrow R$
 $x \rightarrow r$ in radial systems -

Product solutions work for Heisler charts as well!



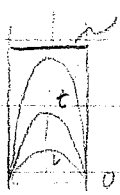
$$\frac{T_m - T_{\infty}}{T_o - T_{\infty}} = \left(\frac{T_m - T_{\infty}}{T_o - T_{\infty}} \right)_x \left(\frac{T_m - T_{\infty}}{T_o - T_{\infty}} \right)_y$$

Can also be applied to 3-D problems.



Recap: Problems have been linear & homogeneous.

1-D Transient

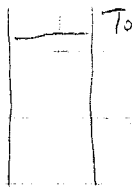


Solved $T(x,t) = X(x)T(t)$

$$T = 2 \sum_{m=0}^{\infty} T_0 (-1)^m \frac{e^{-\alpha \beta_m^2 t}}{\beta_m L} \cos \beta_m x$$

Eigenvalues are from $\cos \beta L = 0$.

$$\beta_m L = (2m+1) \frac{\pi}{2}, \quad m=0, 1, 2, \dots$$



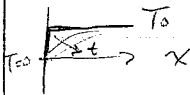
$$h_1 T_0 = 0$$

$$\beta_m L \tan \beta_m L = \frac{hL}{k}$$

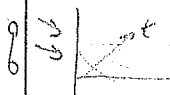
$$\tan \beta_m L = \frac{Bi}{\beta_m L}$$

$$T = \sum_{m=1}^{\infty} C_m e^{-\alpha \beta_m^2 t} \cos \beta_m x$$

Semi-Infinite



$$\frac{T}{T_0} = \text{erf} \left(\frac{x}{\sqrt{4\alpha t}} \right)$$



$$T(x,t) = \frac{T_0}{k} \left[\left(\frac{\alpha t}{\pi} \right)^{1/2} e^{-\frac{x^2}{4\alpha t}} - \frac{x}{2} \text{erfc} \left(\frac{x}{\sqrt{4\alpha t}} \right) \right]$$

$$T_{\text{surface}} \propto t^{1/2}$$

Product Solution

$$y = \frac{1}{x} \cdot \frac{xy}{xy}$$

$$T(x,y,t) = T_1(x,t) \cdot T_2(y,t)$$

$$\text{IC: } T_0 = T_0 \cdot 1$$

or

$$= \sqrt{T_0} \cdot \sqrt{T_0}$$

Works for for homogeneous linear problem only.

$$\nabla^2 T = \frac{1}{x} \frac{\partial T}{\partial x}$$

$$k \frac{\partial T}{\partial n} + hT = 0$$

Reductive BC, T_{∞} will not work.

For T_{∞} , transform into $\theta = T - T_{\infty}$.

Multidimensional Steady State

$$\nabla^2 T = 0$$

$$T=0 \quad \frac{dT}{dx}=0 \quad \boxed{\nabla^2 T=0} \quad k \frac{\partial T}{\partial x} + h T = 0 \quad \leftarrow \text{homogeneous}$$

$$k \frac{\partial T}{\partial n_i} + h_i T = f_i \quad \leftarrow \text{not homogeneous}$$

One nonhomogeneous BC is not a problem

For multiple nonhomogeneous BC's:

$$T = \phi_1 \quad \boxed{\nabla^2 T = 0} \quad T = \phi_2 \quad \boxed{\nabla^2 T = 0} \quad T = \phi_3 \quad \boxed{\nabla^2 T = 0} \quad T = \phi_4 \quad \boxed{\nabla^2 T = 0}$$

$$\text{Solution: } T = T_1 + T_2 + T_3 + T_4$$

Since

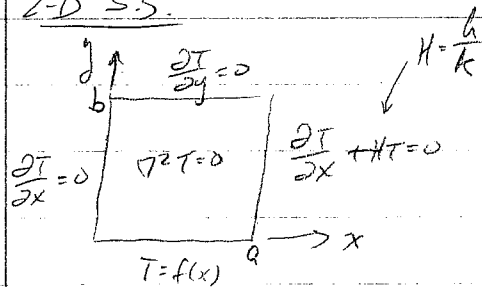
$$\nabla^2 (T_1 + T_2 + T_3 + T_4) = 0$$

General form:

$$k_i \frac{\partial T_j}{\partial n_i} + h_i T_j = \delta_{ij} f_i \quad \text{or } S_i \quad \text{where } \delta_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$T = \sum_j T_j$$

2-D S.S.



$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \quad \text{elliptic.}$$

$$T(x, y) = X(x) Y(y)$$

$$\frac{d^2 X}{dx^2} + \beta^2 X = 0$$

$$\frac{d^2 Y}{dy^2} - \beta^2 Y = 0$$

Minus sign is needed when we look at decays.

Since $f(x)$ is present, we need cos & sin values.

$$\begin{aligned} \frac{dX}{dx} &= 0 \quad @ \quad x=0 \\ \frac{dX}{dx} + HX &= 0 \quad @ \quad x=a \end{aligned}$$

$$\frac{dY}{dy} = 0 \quad @ \quad y=b$$

Solving,

$$X = A \cos \beta x + B \sin \beta x$$

$$X(\beta_n, x) = \cos \beta_n x \quad \& \quad \beta_n \tan \beta_n a = H$$

$$\frac{1}{N(\beta_n)} = \int_0^a X^2 dx = 2 \frac{\beta_n^2 + H^2}{a(\beta_n^2 + H^2) + H}$$

$$\begin{aligned} Y &= A \cosh \beta y + B \sinh \beta y \\ &= \cosh \beta_n (b-y) \end{aligned}$$

$$T(x, y) = \sum_{n=1}^{\infty} C_n \cosh \beta_n (b-y) \cos \beta_n x$$

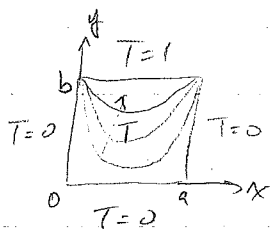
$$f(x) = \sum_{m=1}^{\infty} C_m \cosh \beta_m b \cos \beta_m x$$

Multiply by $\cos \beta_n x$,

$$C_m = \frac{1}{N(\beta_m) \cosh \beta_m b} \int_0^a \cos \beta_m x' f(x') dx'$$

If $f(x) = T_0$,

$$C_m = \frac{T_0}{N(\beta_m) \cosh \beta_m b} \int_0^a \cos \beta_m x' dx' \quad \rightarrow \sin \beta_m a$$



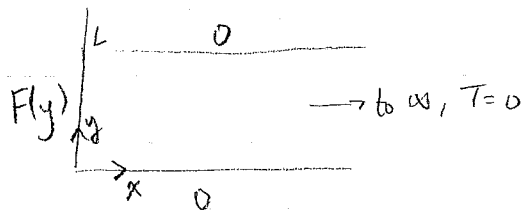
$$T(x,y) \sim \sin \beta x \sinh \beta y$$

$$\sin \beta_m a = 0 \Rightarrow \beta_m = \frac{m\pi}{a}, \quad m=1, 2, 3, \dots$$

$$T = \sum_{m=1}^{\infty} C_m \sin \frac{m\pi x}{a} \sinh \frac{m\pi y}{a}$$

$$1 = \sum_{m=1}^{\infty} C_m \sin \frac{m\pi x}{a} \sinh \frac{m\pi b}{a}$$

$$C_m = \frac{2[(-1)^{m+1} + 1]}{m\pi \sinh(\frac{m\pi b}{a})}, \quad m=1, 2, 3, \dots$$



$$T(x,y) = X(x) Y(y)$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} - \beta^2 X = 0 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} + \beta^2 Y = 0$$

$$X = A_1 e^{\beta x} + A_2 e^{-\beta x} \quad Y = A_3 \cos \beta y + A_4 \sin \beta y$$

$$T(x,y) = \sum C_m e^{-\beta_m x} \sin \beta_m y$$

$$\sin \beta_m L = 0$$

$$\beta_m L = m \pi$$

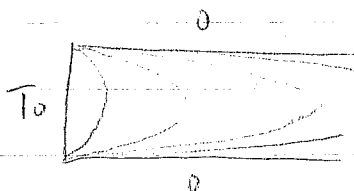
$$\beta_m = \frac{m \pi}{L}$$

If $F(y) = T_0$, we can find C_m

If $F(y) = T_0 \sin \frac{\pi y}{L}$, orthogonality collapses to $m=1$.

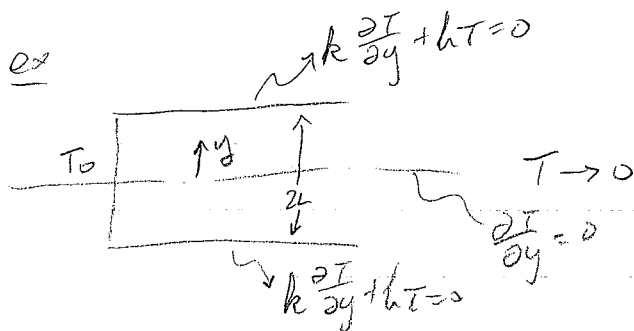
$$T(x,y) = T_0 e^{-\frac{\pi x}{L}} \sin \frac{\pi y}{L}$$

We don't have a series!! $\rightarrow$ Very attractive



$$\frac{ex}{chip}$$

Q2

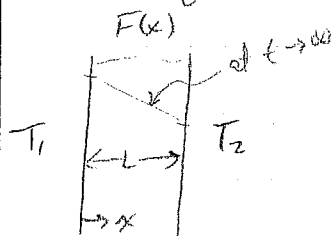


Solve half problem.

$$\tan \beta_n L = \frac{Bi}{\beta_n L}$$

$$T \sim \cos \beta_n y e^{-\beta_n x}$$

Nonhomogeneous Problems



$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$T(x, t) = T_{ss}(x) + T_h(x, t)$$

$$\frac{d^2 T}{dx^2} = 0$$

$$\frac{\partial^2 T_h}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T_h}{\partial t}$$

$$T_s = T_1 @ x=0$$

$$T_h = 0 @ x=0$$

$$T_s = T_2 @ x=L$$

$$T_h = 0 @ x=L$$

$$T_h = F(x) - T_s(x) @ t=0.$$

$$T_s = T_1 + (T_2 - T_1) \frac{x}{L}$$

$$T_h(x, t) = \sum_{n=1}^{\infty} e^{-\alpha \beta_n^2 t} \frac{\int_0^L \beta_n(x') [F(x') - T_s(x')] dx'}{N(\beta_n)}$$

$$\frac{1}{N(\beta_m)} = \frac{2}{L}$$

$$\sin \beta_m L = 0$$

$$X(\beta_m, x) = \sin \beta_m x$$

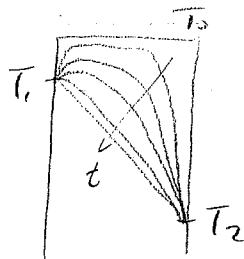
$$T_s = T_1 + (T_2 - T_1) \frac{x}{L}$$

Soln:

$$T(x, t) = T_s + T_h$$

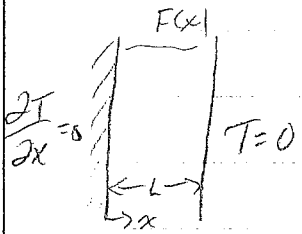
$$T(x, t) = \overset{\text{S.S.}}{T_1 + (T_2 - T_1) \frac{x}{L}} + \frac{2}{L} \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \sin \beta_m x \int_0^L F(x') \sin \beta_m x' dx' \\ + \frac{2}{L} \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \frac{1}{\beta_m} \sin \beta_m x [T_2 \cos m\pi - T_1]$$

where $\cos m\pi = (-1)^m$



Solution allows accurate answers at very small times compared to numerical answers.

General form



$$\frac{\partial^2 T}{\partial x^2} + \frac{q_0}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$T = T_s(x) + T_h(x, t)$$

$$\frac{d^2 T_s}{dx^2} + \frac{q_0}{k} = 0$$

$$\frac{dT_s}{dx} = 0 \text{ @ } x=0$$

$$T_s = 0 \text{ @ } x=L$$

$$\frac{\partial^2 T_h}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T_h}{\partial t}$$

$$\frac{\partial T_h}{\partial x} = 0 \text{ @ } x=0$$

$$T_h = 0 \text{ @ } x=L$$

$$T_h = F(x) - T_s(x) \text{ @ } t=0$$

$$T_s = \frac{q_0 L^2}{2k} \left(1 - \frac{x^2}{L^2}\right)$$

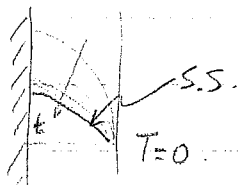
$$T_h = \frac{2}{L} \sum_{m=0}^{\infty} e^{-\alpha \beta_m^2 t} \cos \beta_m x \int_0^L [F(x') - T_s(x')] \cos \beta_m x' dx'$$

$$\cos \beta_m L = 0 \Rightarrow \beta_m = \left(\frac{2m+1}{2L}\right) \pi, \quad m=0, 1, 2, \dots$$

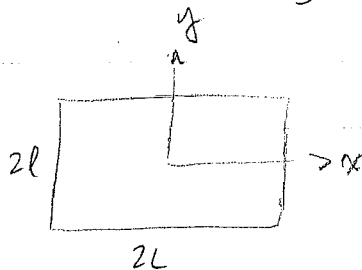
$$T(x, t) = T_s(x) + T_h(x, t)$$

$$T(x, t) = \frac{q_0 L^2}{2k} \left(1 - \frac{x^2}{L^2}\right) + \frac{2}{L} \sum_{m=0}^{\infty} e^{-\alpha \beta_m^2 t} \cos \beta_m x \int_0^L F(x') \cos \beta_m x' dx'$$

$$= \frac{2q_0}{kL} \sum_{m=0}^{\infty} (-1)^m e^{-\alpha \beta_m^2 t} \frac{1}{\beta_m^3} \cos \beta_m x$$



S.S. 2-D with energy generation



non homogeneous

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{q_0}{k} = 0$$

BC's:

$$\frac{\partial T}{\partial x}(0, y) = 0 \quad T(L, y) = 0$$

$$\frac{\partial T}{\partial y}(x, 0) = 0 \quad T(x, L) = 0$$

$$T(x, y) = \phi(x) + \psi(x, y)$$

$$\frac{d^2 \phi}{dx^2} + \frac{q_0}{k} = 0$$

$$\frac{d\phi}{dx} = 0 \text{ @ } x=0$$

$$\phi = 0 \text{ @ } x=L$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\frac{\partial \psi}{\partial x}(0, y) = 0$$

$$\frac{\partial \psi}{\partial y}(x, 0) = 0$$

$$\psi(L, y) = 0$$

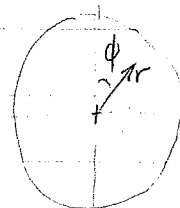
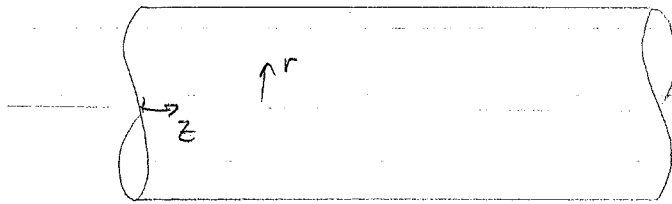
$$\psi(x, L) = -\phi(x)$$

$$\phi = \frac{q_0 L^2}{2k} \left(1 - \frac{x^2}{L^2} \right)$$

$$\frac{T(x, y)}{q_0 L^2 / k} = \frac{1}{2} \left[1 - \left(\frac{x}{L} \right)^2 \right] - 2 \sum_{m=0}^{\infty} \frac{(-1)^m}{(\beta_m L)^3} \frac{\cosh \beta_m y}{\cosh \beta_m L} \cdot \cosh \beta_m x$$

Cylindrical Coordinates

$$\nabla^2 T = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$



$$T(r, \phi, z, t)$$

$$\underbrace{\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r}}_{\text{radial}} + \underbrace{\frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2}}_{\text{circumferential}} + \underbrace{\frac{\partial^2 T}{\partial z^2}}_{\text{axial}} = \underbrace{\frac{1}{\alpha} \frac{\partial T}{\partial t}}_{\text{transient}}$$

$$T(r, \phi, z, t) = \psi(r, \phi, z) \Gamma(t)$$

$$\frac{1}{\psi} \left(\frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2} \right) = \frac{1}{\Gamma} \frac{d\Gamma}{dt} = -\lambda^2$$

$$\Gamma = e^{-\alpha \lambda^2 t}$$

$$\psi(r, \phi, z) = R(r) \Phi(\phi) Z(z)$$

$$\frac{1}{R} \left(\frac{d^2 R}{dr^2} + \frac{1}{r} \frac{dR}{dr} \right) + \underbrace{\frac{1}{r^2} \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2}}_{-\nu^2} + \underbrace{\frac{1}{Z} \frac{d^2 Z}{dz^2}}_{-\eta^2} = -\lambda^2$$

$$\frac{d^2 Z}{dz^2} + \eta^2 Z = 0$$

$$\rightarrow \sin \eta z, \cos \eta z$$

$$\frac{d^2 \Phi}{d\phi^2} + \nu^2 \Phi = 0$$

$$\rightarrow \sin \nu \phi, \cos \nu \phi$$

$\nwarrow R \text{ depends on } \nu$

$$\frac{1}{R_\nu} \left(\frac{d^2 R_\nu}{dr^2} + \frac{1}{r} \frac{dR_\nu}{dr} \right) - \frac{\nu^2}{r^2} = -\beta^2$$

$$\frac{d^2 R_\nu}{dr^2} + \frac{1}{r} \frac{dR_\nu}{dr} + \left(\beta^2 - \frac{\nu^2}{r^2} \right) R_\nu = 0$$

If $\nu = 0$, R_0 .

If $\nu \neq 0$, R_ν .

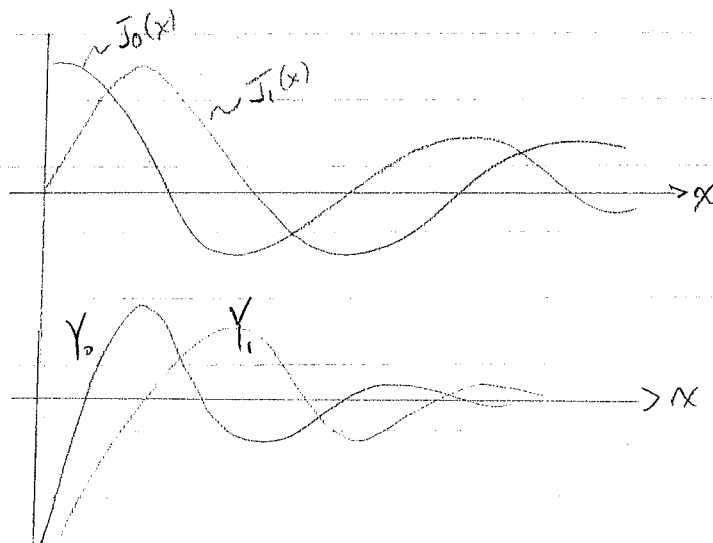
This is known as Bessel's Equation.

$$\lambda^2 = \beta^2 + \eta^2$$

Solution of Bessel's Eqn. is

$$R_\nu(\beta, r) = A_1 J_\nu(\beta r) + A_2 Y_\nu(\beta r)$$

Where $J_\nu(\beta r)$ & $Y_\nu(\beta r)$ are Bessel Functions of the 1st and 2nd kind



$$J_0' = -J_1$$

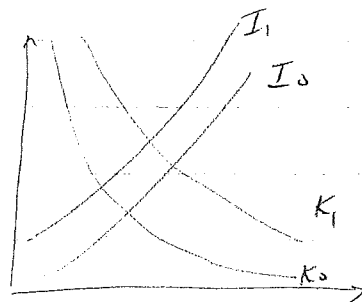
$$\frac{d^2 R_\nu}{dr^2} + \frac{1}{r} \frac{dR_\nu}{dr} - \left(\beta^2 + \frac{\nu^2}{r^2} \right) R_\nu = 0$$

↙ minus

We get modified Bessel Functions

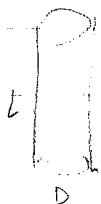
$$I_\nu(\beta r) = J_\nu(i\beta r), \quad i = \sqrt{-1}$$

$$K_\nu(\beta r) = Y_\nu(i\beta r)$$



$$I_0' = I_1$$

$$K_0' = -K_1$$



No ϕ Dependence

$$\frac{d^2 \underline{z}}{dz^2} + \eta^2 \underline{z} = 0$$

$$\rightarrow \sin \eta z, \cos \eta z$$

$$\frac{d^2 R_0}{dr^2} + \frac{1}{r} \frac{dR_0}{dr} + \beta^2 R_0 = 0 \rightarrow J_0(\beta r), Y_0(\beta r)$$

$$P = e^{-\alpha \lambda^2 t}, \quad \lambda^2 = \beta^2 + \eta^2$$

No z Dependence

$$\frac{d^2 \underline{\Phi}}{d\phi^2} + \nu^2 \underline{\Phi} = 0 \rightarrow \cos \nu \phi, \sin \nu \phi$$

$$\frac{d^2 R_\nu}{dr^2} + \frac{1}{r} \frac{dR_\nu}{dr} + \left(\beta^2 - \frac{\nu^2}{r^2} \right) R_\nu = 0 \rightarrow J_\nu(\beta r), Y_\nu(\beta r)$$

$$P = e^{-\alpha \lambda^2 t}, \quad \lambda^2 = \beta^2$$

No t Dependence (S.S.)

$$\frac{d^2 \underline{\Phi}}{d\phi^2} + \nu^2 \underline{\Phi} = 0 \rightarrow \sin \nu \phi, \cos \nu \phi$$

$$\frac{d^2 \underline{z}}{dz^2} + \eta^2 \underline{z} = 0 \rightarrow \sin \eta z, \cos \eta z$$

$$\frac{d^2 R_\nu}{dr^2} + \frac{1}{r} \frac{dR_\nu}{dr} - \left(\eta^2 + \frac{\nu^2}{r^2} \right) R_\nu = 0 \rightarrow I_\nu(\eta r), K_\nu(\eta r)$$

$\sim$ depends on how η is chosen

choose $-\eta^2$
if cylinder is
semi-infinite

No z, t Dependence

$$\frac{d^2 \Phi}{dr^2} + V^2 \Phi = 0$$

$$\frac{d^2 R}{dr^2} + \frac{1}{r} \frac{dR}{dr} - \frac{V^2}{r^2} R = 0 \rightarrow \begin{cases} r^V, r^{-V} \text{ if } V \neq 0 \\ C_1 + C_2 \ln r \text{ if } V = 0 \end{cases}$$

1-D transient



$$\text{IC: } T = F(r) @ t = 0$$

$$\text{BC: } \frac{\partial T}{\partial r} + HT = 0 @ r = b$$

$$\frac{\partial T}{\partial r} = 0 @ r = 0.$$

$$T = e^{-\alpha \beta_m^2 t}, \quad V = 0$$

$$\frac{d^2 R_0}{dr^2} + \frac{1}{r} \frac{dR_0}{dr} + \beta^2 R_0 = 0 \rightarrow J_0(\beta r), Y_0(\beta r) \quad \text{finite @ } r \rightarrow 0.$$

$$T(r, t) = \sum_{m=1}^{\infty} C_m e^{-\alpha \beta_m^2 t} R_0(\beta_m, r)$$

$$\text{IC: } F(r) = \sum_{m=1}^{\infty} C_m R_0(\beta_m, r) \rightarrow \text{gives } C_m \text{'s.}$$

Orthogonality:

$$\int_0^b r R_\nu(\beta_m, r) R_\nu(\beta_n, r) dr = \begin{cases} 0 & \text{if } m \neq n \\ N(\beta_m) & \text{if } m = n \end{cases}$$

$$N(\beta_m) = \int_0^b r R_\nu^2 dr$$

$$C_m = \frac{1}{N(\beta_m)} \int_0^b r R_0 F(r) dr$$

$$R_0 \sim J_0(\beta_m r) \quad \text{since } Y_0(\beta_m r) \rightarrow \infty \text{ and is dropped.}$$

BC at $r=b$:

$$\beta_m J_0'(\beta_m b) + H J_0(\beta_m b) = 0.$$

$$\boxed{\beta_m J_1(\beta_m b) = H J_0(\beta_m b)} \rightarrow \text{gives } \beta_m$$

Solution is

$$T(r, t) = \sum_{m=1}^{\infty} \frac{1}{N(\beta_m)} e^{-\alpha \beta_m^2 t} R_0(\beta_m r) \int_0^b r' R_0(\beta_m r') F(r') dr'$$

$$\boxed{T(r, t) = \frac{2}{b^2} \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \frac{\beta_m^2 J_0(\beta_m r)}{(\beta_m^2 + H^2) J_0(\beta_m b)} \int_0^b r' J_0(\beta_m r') F(r') dr'}$$

$$\text{If } F(r) = T_0,$$

$$T(r, t) = \frac{2HT_0}{b} \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \frac{J_0(\beta_m r)}{(\beta_m^2 + H^2) J_0(\beta_m b)}$$

